
ASSIGNMENT

PROGRAMMING USING C

WEEK – 14

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DEPT & SEC :ECE F

Week – 14

You are transporting some boxes through a tunnel, where each box is a parallelepiped, and is characterized by its length, width and height.

The height of the tunnel **41** feet and the width can be assumed to be infinite. A box can be carried through the tunnel only if its height is strictly less than the tunnel's height. Find the volume of each box that can be successfully transported to the other end of the tunnel. Note: Boxes cannot be rotated.

```
#include<stdio.h>
int main()
{
    int n;
    scanf("%d",&n);
    for(int i=0;i<n;i++)
    {
        int lenght,width,height;
        scanf("%d %d %d",&lenght,&width,&height);

        if(height < 41)
        {
            int volume=lenght*width*height;
            printf("%d\n",volume);
        }
    }
}
```

	Input	Expected	Got	
✓	4 5 5 5 1 2 40 10 5 41 7 2 42	125 80	125 80	✓

Passed all tests! ✓

You are given n triangles, specifically, their sides a_i , b_i and c_i . Print them in the same style but sorted by their areas from the smallest one to the largest one. It is guaranteed that all the areas are different.

```
#include<stdio.h>
#include<math.h>
#include<stdlib.h>
typedef struct
{
    double area;
    int a,b,c;
}Triangle;

double calculate_area(int a, int b, int c)
{
    double p=(a+b+c)/2.0;
    return sqrt(p*(p-a)*(p-b)*(p-c));
}
int compare(const void*x,const void*y)
{
    Triangle *t1=(Triangle *)x;
    Triangle *t2=(Triangle *)y;
    if (t1->area < t2->area)return -1;
    if (t1->area > t2->area)return 1;
    return 0;
}
int main()
{
    int n;
    scanf("%d",&n);
    Triangle triangles[n];

    for (int i=0; i<n;i++)
    {
        int a,b,c;
        scanf("%d %d %d",&a,&b,&c);

        triangles[i].a = a;
        triangles[i].b = b;
        triangles[i].c = c;
        triangles[i].area = calculate_area(a,b,c);
    }

    qsort(triangles, n, sizeof(Triangle),compare);

    for(int i=0;i<n;i++){
        printf("%d %d %d\n",triangles[i].a, triangles[i].b, triangles[i].c);
    }
    return 0;
}
```

	Input	Expected	Got	
✗	3 7 24 25 5 12 13 3 4 5	3 4 5 5 12 13 7 24 25	7 24 25 5 12 13 3 4 5	✗

Some hidden test cases failed, too.

Your code must pass all tests to earn any marks. Try again.

Show differences